



DF-8805 Datasheet

Digital Force Sensor

FEATURES

- Integrated Analog Front-End
- Integrated Instrumentation Amplifier with Programmable Gain
- Auto Preload Adjustment
- Ultra-Low Power
- Low Latency
- Programmable Temperature Correction
- Software- and Hardware-Programmable Interrupt Threshold
- Industry-Standard I²C Interface
- Ultra-Small 6-pin WLCSP

Description

The DF-8805 is an industry-leading single-chip digital force sensor with integrated analog front-end, ultra-low power, and high SNR.

The integrated digital output provides seamless connection to host processors via a standard I²C interface, simplifying system integration while reducing external component count and PCB area requirements.

The sensor is designed for Always-On applications, this architecture minimizes host processor wake events, enabling ultra-low system power consumption by maintaining the host in low-power states until a predefined force threshold is exceeded. A dual-function threshold programming pin (THR_ADR) allows external resistor configuration of trigger thresholds.

The sensor has two modes of operation, a 12-bit ultra-low power mode and a 16-bit high performance mode. The sensor incorporates automatic preload adjustment for offset correction to compensate for large preload changes caused by assembly processes and ambient temperature variations. Programmable temperature correction in conjunction with an integrated temperature sensor provides additional compensation for offset changes caused by ambient temperature.

The sensor is compatible with standard surface-mount assembly processes, including pick-and-place equipment and reflow soldering profiles, facilitating high-volume manufacturing on both rigid PCB and flexible FPCB substrates.

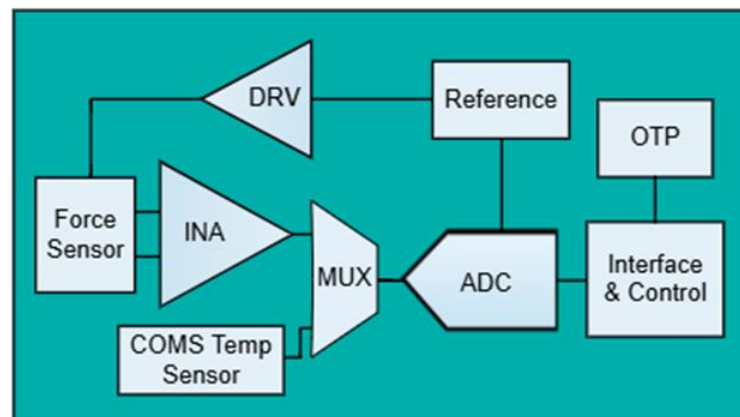
Applications

- Automotive
- Consumer: Smartphones, Tablets, Wearables
- IoT: Smart Home Devices
- Input Devices: Stylus, Touchpads
- Gaming and AR/VR: Force-Sensing Controls

Characteristics

- 3.3 V and 1.8 V Supply Voltage
- 2 μ A @ 15 Hz
- Package Size: 1.33 x 1.33 x 0.25 mm³
- Minimum Latency: 1 ms

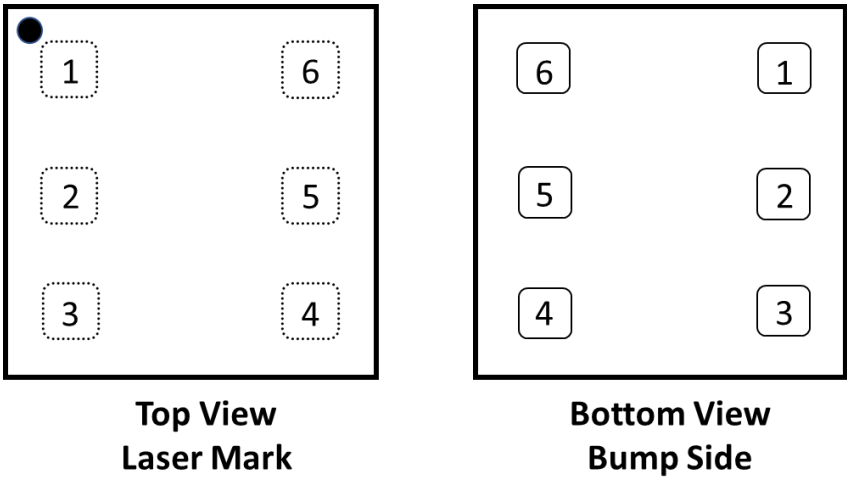
Functional Block Diagram



Ordering Information

PART NUMBER	OPERATING TEMPERATURE	GRADE	DESCRIPTION
DF-8805B00BSP	-40 °C to +85 °C	Automotive Grade 3	5000 pc 7-inch reel (with backside coating)
DF-8805B00SRBSP	-40 °C to +85 °C	Automotive Grade 3	300 pc sample reel (with backside coating)
DF-8805B00SQBSP	-40 °C to +85 °C	Automotive Grade 3	25 pc sample bag (with backside coating)
DF-8805C00BSP	-40 °C to +105 °C	Automotive Grade 2	5000 pc 7-inch reel (with backside coating)
DF-8805C00SRBSP	-40 °C to +105 °C	Automotive Grade 2	300 pc sample reel (with backside coating)
DF-8805C00SQBSP	-40 °C to +105 °C	Automotive Grade 2	25 pc sample bag (with backside coating)
DF-8800DK	N/A	N/A	Full Design Kit
DF-8800PCK	N/A	N/A	Design Kit, Sensor Module only

Pin Configuration



Pin Description

SOLDER BUMP	PIN NAME	PIN DESCRIPTION
1	THR_ADR	Threshold/Address Pin. Dual function: I ² C address selection and hardware-programmed force threshold setting at power-up.
2	INTB	Interrupt Output. Alert the host when programmed force is exceeded.
3	SCL	Serial Clock Input. I ² C clock input.
4	SDA	Serial Data I/O. Bidirectional I ² C data line.
5	VDD	Power Supply.
6	GND	Ground. Common ground reference for all signals.

Absolute Maximum Ratings

Stresses beyond those listed may cause permanent damage to the device. These are stress ratings only; functional operation of the device under these or any other conditions beyond those indicated in the operational sections is not implied. Prolonged exposure to absolute maximum rating conditions may degrade device reliability.

PARAMETER	VALUE
VDD to GND (Power)	-0.3 V to +4.0 V
THR_ADR L, SDA to GND (Digital Input/Output)	-0.3 V to +4.0 V
INTB to GND (Digital Output)	-0.3 V to +4.0 V
SCL to GND (Digital Input)	-0.3 V to +4.0 V
Applied Force (directly applied on top of the sensor) ¹	20 N
Applied Force (applied at gauge mode) ²	40 N
Operating Temperature Range	See Ordering Information
Storage Temperature	-55 °C to +150 °C
Lead Temperature (Soldering 10 s)	+260 °C
Electrostatic Discharge Protection (ESD)	2000 V (HBM), 500 V (CDM)

¹ Not an operation mode, values based on characterization at room temperature using a 1.2 mm thick 4-layer FR4 material, 10 mm flat tip with 0.4 mm thick 75A rubber.

² Values based on characterization at room temperature using a 1.2 mm thick 4-layer FR4 material with 8 mm diameter cavity, rounded metal test tip of 2.65 mm diameter and 4 mm radius. At gauge mode, force is applied to the PCB side.

Electrical Characteristics

VDD = 1.80 V ± 10% or 3.30 V ± 10%, T_A = operating temperature range of the device, Typical values are at T_A = +25 °C. Min/max limits are 100% tested at room temperature unless otherwise noted.

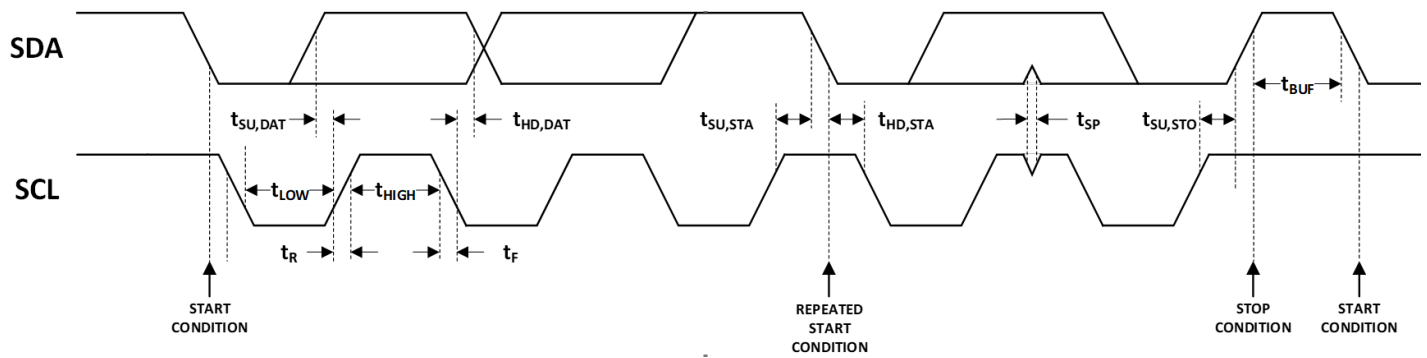
PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP.	MAX.	UNITS
ANALOG-TO-DIGITAL CONVERTER						
Resolution		Ultra-Low Power (ULP) Mode		12		Bits
		High Performance (HP) Mode		16		
Maximum Data Rate	F _S	WAIT = 000	ULP Mode	1000		Sample/s
			HP Mode, 4-order	250		
			HP Mode, 16-order	63		
POWER SUPPLY REQUIREMENTS						
Power Supply	VDD		1.62		3.63	V
Supply Current	I _{CC}	WAIT = 000, ULP Mode		60	120	μA
		WAIT = 011, ULP Mode		8.5		
		WAIT = 000, HP Mode, 16-order		90		
		WAIT = 011, HP Mode, 16-order		62		
Shutdown Current	I _{SHDN}	EN=0	VDD = 1.8V	0.7	1.5	μA
			VDD = 3.3V	1.0	1.5	
TEMPERATURE SENSOR						
Resolution				0.0625		°C/LSB

PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP.	MAX.	UNITS
FORCE SENSOR_ULP MODE (INAGAIN=011 x31, ADCRAW=1, PRELDADJ =0)						
Sensitivity ¹	S _V		69	133	197	LSB/N
Temperature Coefficient of Sensitivity ^{1, 2}	TCS		-1	0	1	LSB/N/°C
Offset ^{1, 2}	V _{OS}			-872		LSB
Temperature Coefficient of Offset ^{1, 3}	TCO		-10	9	28	LSB/°C
Linearity ¹		Best straight-line fit	0.98		1.0	
Noise				0.5		LSB _{RMS}
FORCE SENSOR_HP MODE (INAGAIN=011 x31, ADCRAW=1, PRELDADJ =0)						
Sensitivity ¹	S _V		1100	2130	3150	LSB/N
Temperature Coefficient of Sensitivity ^{1, 2}	TCS		-16	0	16	LSB/N/°C
Offset ^{1, 2}	V _{OS}			-13950		LSB
Temperature Coefficient of Offset ^{1, 3}	TCO		-160	136	450	LSB/°C
Linearity ¹		Best straight-line fit	0.98		1.0	
Noise		4-order		6		LSB _{RMS}
		16-order		4		
DIGITAL INTERFACE						
Output Low Voltage	V _{OL}	VDD = 1.8 V, I _{SINK} = 2 mA			0.36	V
		VDD = 3.3 V, I _{SINK} = 6 mA			0.4	
I ² C Input Voltage, High	V _{IH}	SDA, SCL	0.7 * VDD			V
I ² C Input Voltage, Low	V _{IL}	SDA, SCL			0.3 * VDD	V
Input Hysteresis	V _{HYS}			270		mV
Input Capacitance	C _{IN}			1		pF
Input Leakage Current	I _{IN}	VDD = 0 V & VDD=3.63 V			1	μA
I ² C TIMING CHARACTERISTICS ³						
Serial Clock Frequency	f _{SCL}		100		400	kHz
Bus Free Time Between STOP and START	t _{BUF}		1.3			μs
Hold Time (Repeated) START Condition	t _{HD,STA}		0.6			μs
Low Period of the SCL Clock	t _{LOW}		1.3			μs
High Period of the SCL Clock	t _{HIGH}		0.6			μs
Setup Time for a Repeated START	t _{SU,STA}		0.6			μs
Setup Time for STOP Condition	t _{SU,STO}		0.6			μs
Data Hold Time	t _{HD,DAT}		0			μs
Data Setup Time	t _{SU,DAT}		100			ns
Bus Capacitance	C _B	f _{SCL} =400kHz			400	pF
SDA and SCL Rise Time	t _R		20		300	ns
SDA and SCL Fall Time	t _F		20*VDD/5.5		300	ns
Pulse Width of Suppressed Spike	t _{SP}		0		50	ns
I ² C Timeout	t _{TO}			51.2		ms

¹ Tested at gauge mode, force is applied to the PCB side. Values based on characterization at room temperature at 3.3V using a 1.2mm FR-4 PCB thickness with 8 mm diameter cavity. Rounded metal test tip in 2.65 mm diameter and 4 mm radius.

² Offset and TCO are measured over temperature in raw mode with baselining turned off. With Baselining turned on, Offset is zero-ed out and baseline tracks the change with temperature.

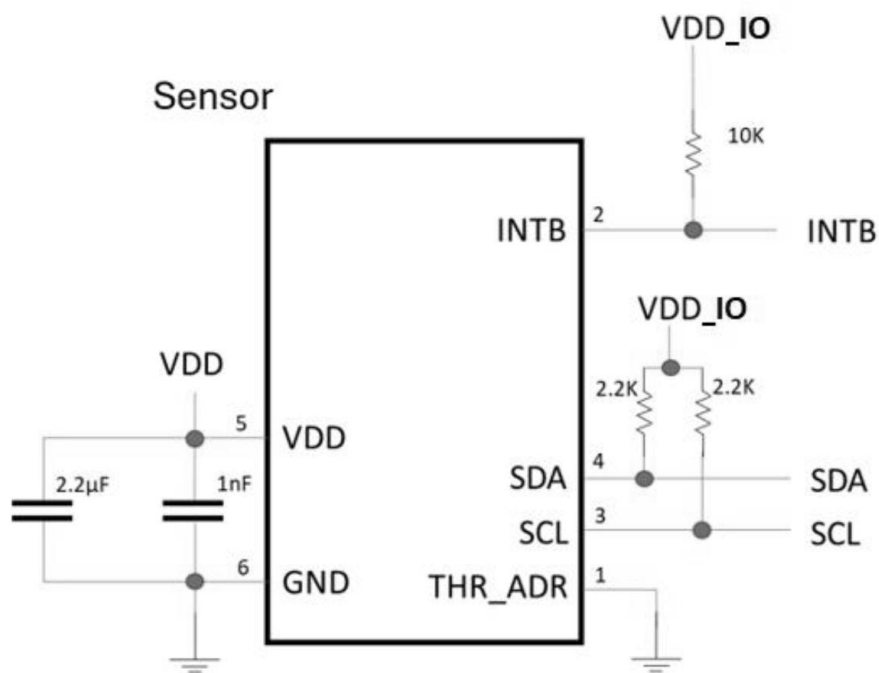
³ Value guaranteed by design.



I²C Communications Timing Diagram

Typical Application Circuit

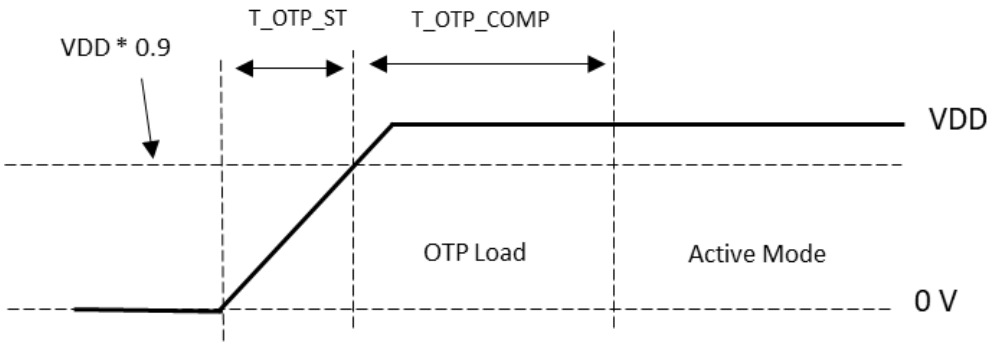
Figure below shows a typical circuit for a button application utilizing Semtech sensor. The sensor generates an interrupt when the programmed threshold is exceeded. The host processor can change the threshold by writing to the sensor registers using the I²C interface.



Power-up Sequence

The sensor will enter active mode by default and be fully operational 10 ms after power is initially applied to the chip. VDD must meet 90% of nominal value within 2 ms after power-up for correct operation, then follow by OTP load in the next 8 ms.

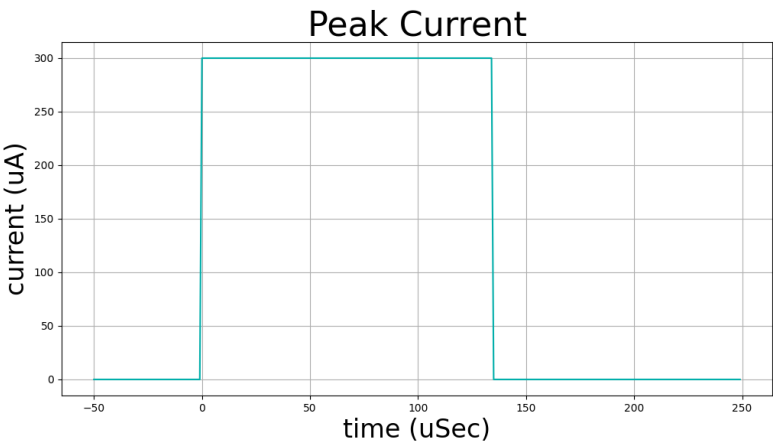
REQUIREMENT	DESCRIPTION	MIN.	TYP.	MAX.	UNITS
T_OTP_ST	Time from power on to OTP load start			2	ms
T_OTP_COMP	Time from OTP load start to complete		8		ms



Power-up Ramp and Full Operation

Peak Current

1x/4x/16x peak current pulses (ULP mode / 4-order HP mode / 16-order HP mode) repeat at the WAIT interval, draws 300 μ A (typ.) for $\sim$ 150 μ s. These pulses do not occur simultaneously and exclude I²C current. Contact Semtech sales for a detailed application note.



Bidirectional Digital Communication Interface

The sensor implements a standard I²C peripheral interface supporting both standard-mode (100 kHz) and fast-mode (400 kHz) operation. The 7-bit I²C address is configured at power-up via the THR_ADR pin, allowing up to 4 sensors on a single bus. Four unique addresses are selectable based on THR_ADR pin connection; see the table below for details.

THR_ADR CONNECTION	SENSOR I ² C ADDRESS ¹
THR_ADR = 3.3 kΩ - 50 kΩ Resistor ² to GND	0x4C
THR_ADR = 3.3 kΩ - 50 kΩ Resistor ² to VDD	0x4D
THR_ADR = GND	0x4E
THR_ADR = VDD	0x4F

¹ 10-bit I²C peripheral addressing and General-Call (broadcast) addressing are not supported.

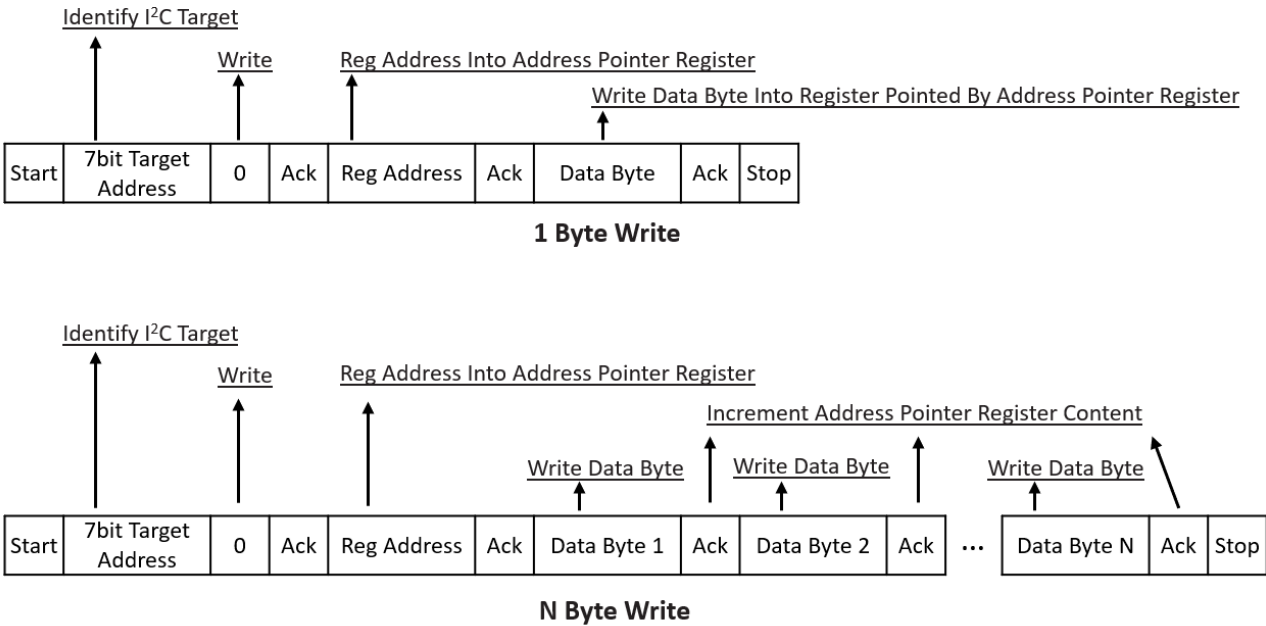
² Use a 10 kΩ resistor if used for address assignment only.

I²C Address Selection using THR_ADR

The sensor has an optional I²C timeout, which resets the bus if either the SCL or SDA signal line is low for ≥ 51 ms. It is recommended that the I²C interface is operated at speeds exceeding 100 kHz to ensure timely reading of the data for low latency sensor output processing. Latency is calculated as INTRSAMPLES x (WAIT + 1) ms at ULP mode.

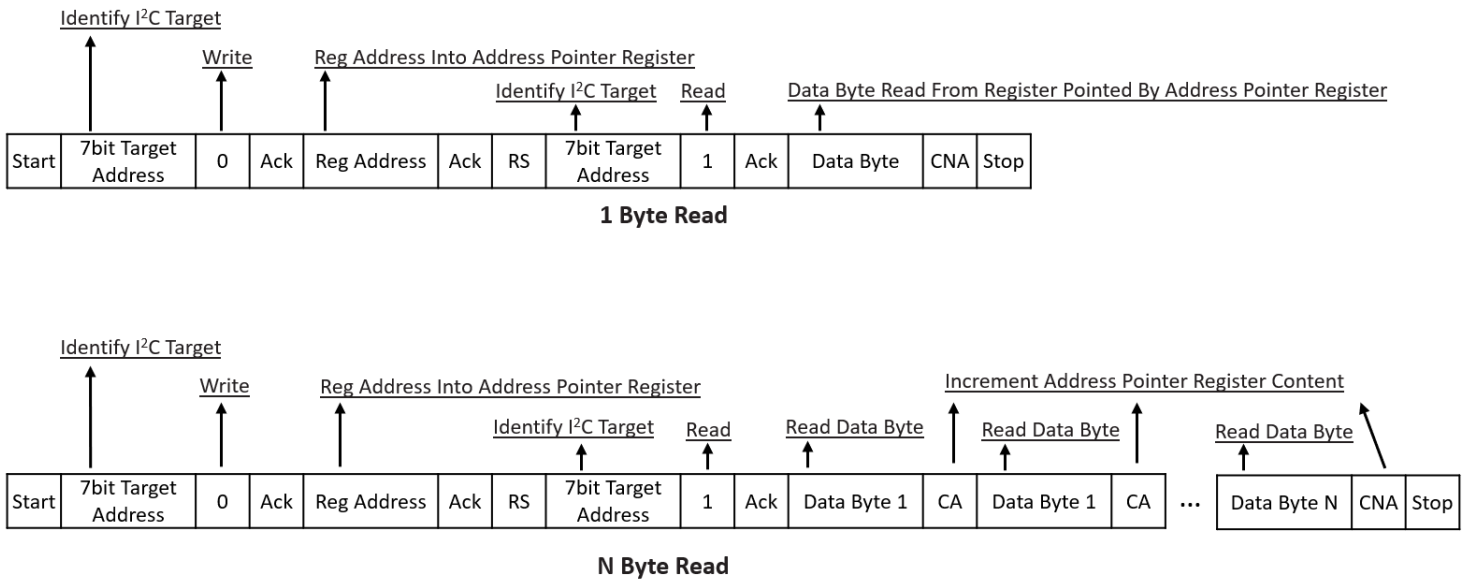
I²C bus signals (SDA and SCL) are open-drain and require external pull-up resistors per I²C specification. This is typically done with a 2 kΩ pull-up resistors (or greater for lower power operation). Series resistors in the signal path are optional to protect the sensor input architecture from high-voltage spikes on the bus lines, minimize crosstalk, and undershoot events of the bus signals. The supply voltage to the sensor should be less than or equal to the pull-up voltage for the I²C and INTB.

The I²C controller initiates all data transfer and generates the SCL clock signal. The sensor responds as a target device, supporting bidirectional communication on the SDA line. The sensor also supports single-byte and multi-byte burst transfers. Figure below shows the frame format for single-byte and multi-byte write and read transfers to and from the sensor.



- Start Start Condition From Controller
- Ack Acknowledgement Bit
- Stop Stop Condition From Controller

I²C Write



RS	Repeated Start Condition From Controller
CA	Controller Acknowledgement
CNA	Controller Not Acknowledgement

I²C Read

Interrupt System

The INTB output enables autonomous sensor operation with minimal host intervention. The interrupt system supports flexible triggering conditions, debounce filtering, and configurable signal polarity to accommodate various interrupt controller architectures.

Interrupt Signal and Operation

Interrupt output is open-drain and requires external pull-up resistor (10 kΩ or greater for low-power operation) to VDD. Interrupt polarity is configurable via INTRPOL register (0x09[3]). Default is active-low.

Interrupt Modes

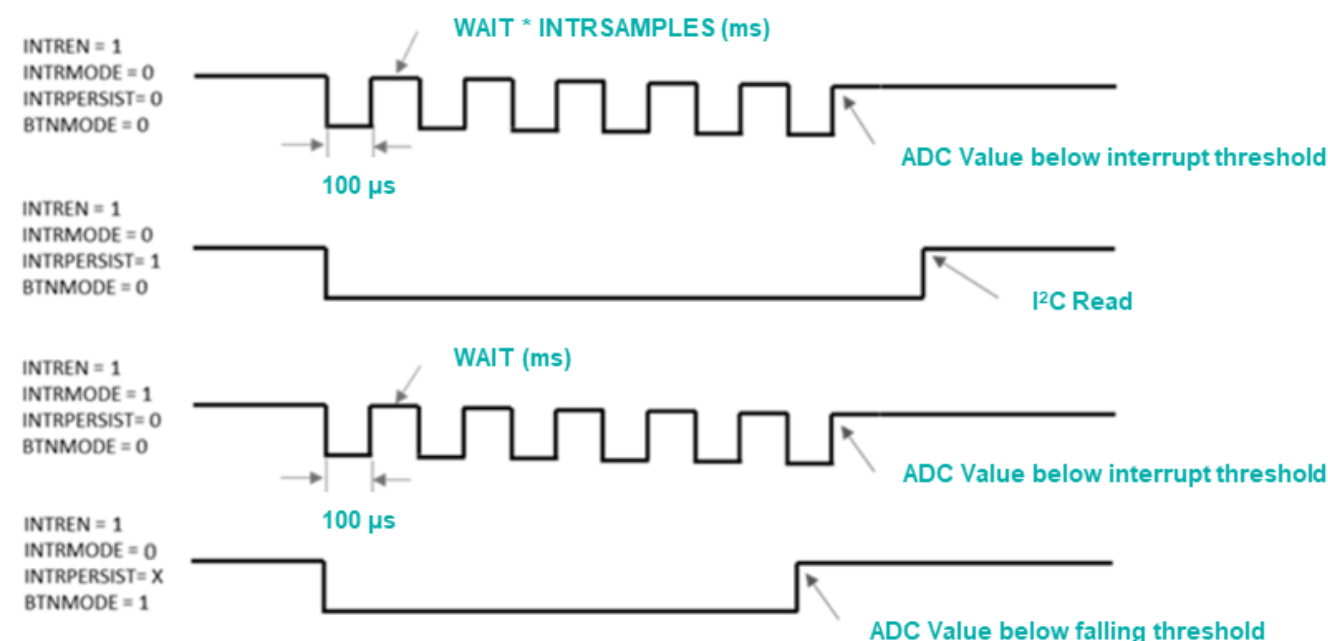
The sensor has configurable interrupt modes (register addresses 0x0C-0x0E and 0x16-0x17), allowing different trigger conditions. Interrupt is generated when the sensor measurement exceeds the programmed threshold set by INTRTHR registers (0x0C-0x0D) for a specified number of consecutive samples set by INTRMODE and INTRSAMPLES.

Standard Mode (BTNMODE = 0):

INTB can be configured to either output 100 μs pulses (INTRPERSIST=0) for as long as the signal is above the threshold level or a constant level (INTRPERSIST=1) until the interrupt status register (0x02) or register 0x01 is read by controller.

Button Mode (BTNMODE = 1 and INTRMODE=0):

Implements press-and-release detection with separate rising set by INTRTHR registers (0x0C-0x0D) and falling thresholds set by FALLTRTHR registers (0x16-0x17), providing hysteresis for stable button emulation.



Interrupt Modes

Hardware Programmable Interrupt Threshold

The sensor implements autonomous interrupt threshold configuration through an external resistor between THR_ADR pin and VDD or GND. This dual-function pin configures both the sensor I²C address and the sensor interrupt threshold (0x0C-0x0D) during power-on reset. The interrupt threshold can be overwritten by the controller if required.

The relationship between external resistor value and programmed threshold follows the empirically derived transfer function:

$$\text{External R [in K}\Omega\text{]} = 1047 / \text{INT threshold [in counts]} - 0.8$$

1% E96 STANDARD RESISTOR VALUE (K Ω)	CALCULATED INT THRESHOLD (COUNTS)
3.3	255
10	97
30.1	34
49.9	21

12-Bit and 16-Bit Modes

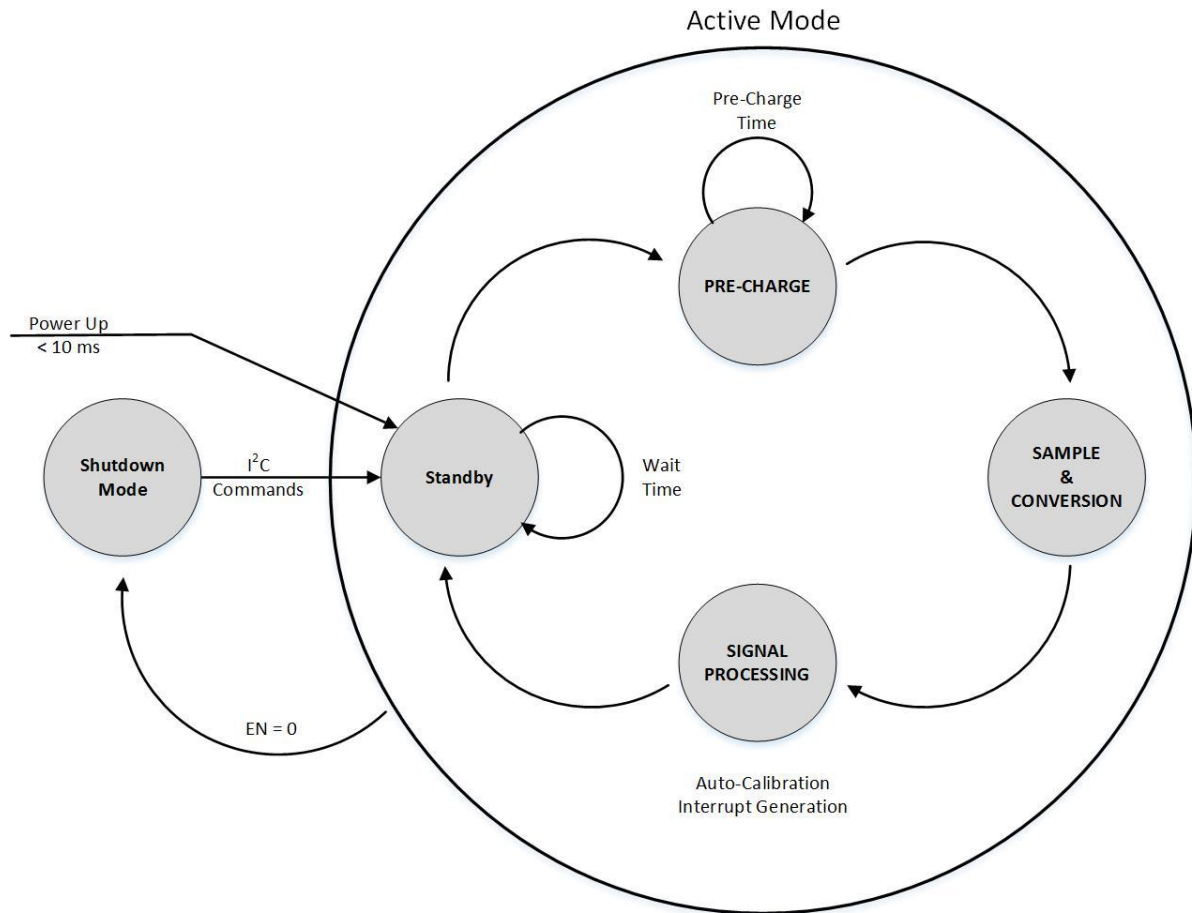
The sensor has two data output types selected by the ENFILTER register: Ultra-Low Power mode (ULP mode) and High-Performance mode (HP mode).

In ULP mode (ENFILTER = 0), the sensor output is 12-bit 2's complement which can be read from ADCOUT[15:4]. The data rate is determined by the WAIT register. Data is updated every "WAIT + 1ms".

In HP mode (ENFILTER = 1), the sensor output is 16-bit 2's complement which can be read from ADCOUT[15:0]. In this mode, there are two settings selected by HIGHORDER register. When HIGHORDER = 0, the device is set to 4-point sum. The data is updated every "WAIT + 4ms". When HIGHORDER = 1, the device is set to 16-point sum. The data is updated every "WAIT + 16ms".

States and Modes of Operation

The sensor implements a state machine architecture with two primary operating modes—Active Mode and Shutdown Mode—optimized for ultra-low-power operation in battery-powered and Always-On applications.



Shutdown Mode

Shutdown mode provides maximum power savings by completely disabling the analog front-end while retaining digital register contents (except PRELDADJ register) and I²C interface responsiveness, allowing the host to read configuration or initiate return to Active Mode via register write.

In this mode, PRELDADJ changes back to POR value when AUTOPRELDADJ is on. If AUTOPRELDADJ is off the value of PRELDADJ will be the last manually-written-in value or the POR, if it is never manually written.

To enter shutdown mode from active mode, disable sensor by writing a "0" to EN register.

When exiting shutdown mode (setting EN from 0 to 1), clear the INTREN and BTNMODE bits. Set both bits to zero before enabling the AFE. If needed, re-enable these bits no earlier than $\Delta t = 3 * (WAIT + PRECHARGE + 1)$ after setting EN to 1.

Active Mode

In active mode, the sensor performs periodic ADC conversions at specified intervals: WAIT+1 ms for ULP mode, WAIT+4 ms for 4-order HP mode, WAIT+16 ms for 16-order HP mode. Between conversions, the analog front-end enters a standby state where all analog circuitry powers down while retaining digital register contents and I²C interface responsiveness.

Baselined Data Output (ADCRAW=0, Default)

This mode implements automatic baseline tracking with continuous offset correction, suitable for applications requiring long-term stability, autonomous interrupt generation, and minimal host intervention.

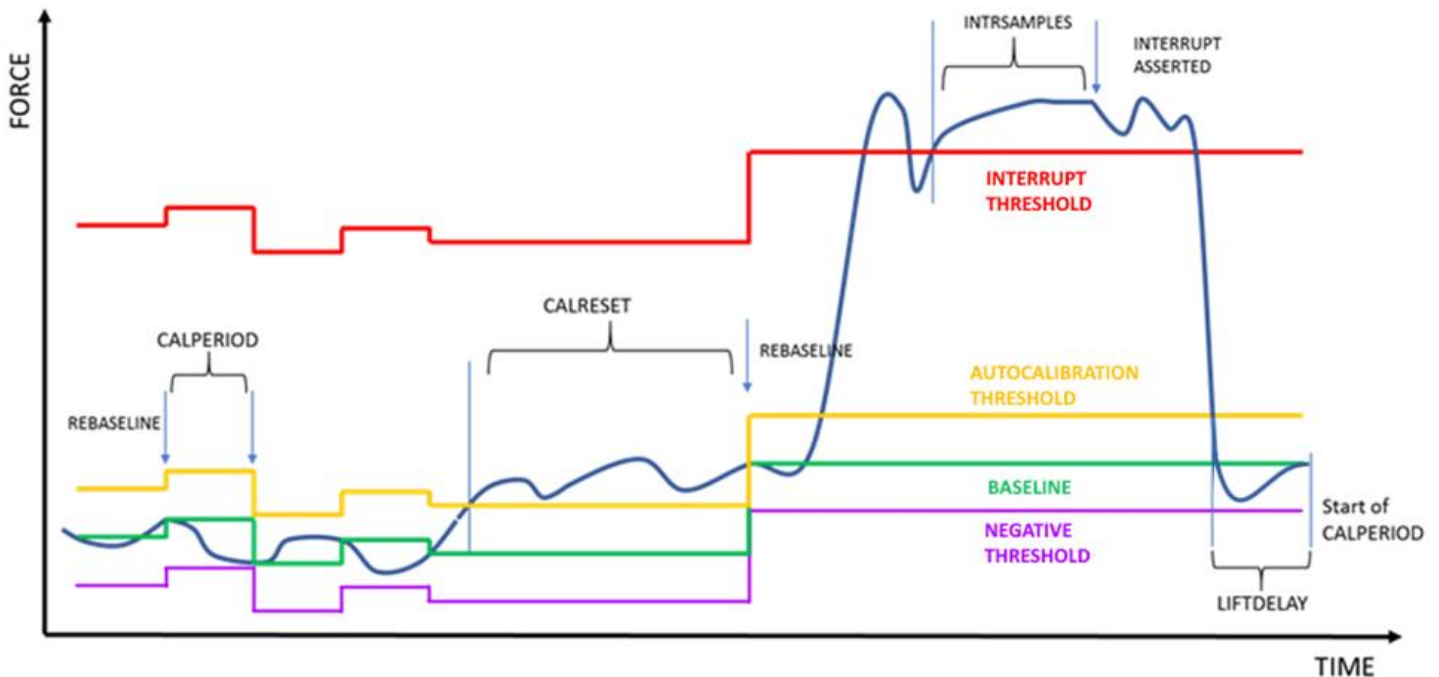
In this mode, the output represents instantaneous force magnitude relative to a dynamically updated zero-force reference point (baseline): $ADCOUT = Raw_ADC - BASELINE$ (in two's complement).

AUTOCAL and INTRTHR thresholds are referenced to current BASELINE value. Interrupt and auto-calibration threshold values shall be positive and interrupt threshold shall be higher than auto-calibration threshold value.

When the sensor signal is less than the AUTOCAL threshold, the baseline is updated at a rate specified by CALPERIOD and weight specified by RISEBLWGT and FALLBLWGT.

When the sensor signal exceeds the auto-calibration threshold, the baseline is updated at a rate specified by CALRESET and weight specified by RESETWGT. There is also an optional negative-side threshold (enabled with ENNEGZONE and specified by NEGTHSEL as a fraction of the AUTOCAL threshold), below which the baseline is only updated after a NEGRESET timeout, again using RESETWGT to calculate the new baseline. The auto-calibration and interrupt levels are referenced to the baseline value; therefore, baseline updates automatically change the stated thresholds. When the interrupt threshold is exceeded for the number of samples set by the INTRSAMPLES register (0x0E), an interrupt is asserted.

When the sensor signal is higher than the interrupt threshold, no auto-calibration is performed unless ENCALMODE is enabled. Once sensor measurements drop below the auto-calibration threshold, the auto-calibration routine is re-enabled after a one-time delay defined by the LIFTDELAY register (0x0A).



Baselining, Auto-calibration & Interrupt Generation

Raw Data Output (ADCRAW=1)

Raw mode disables automatic baseline tracking and automatic offset correction, providing unprocessed ADC data for applications requiring custom signal processing algorithms or where baseline subtraction is performed in host software. This mode is essential for advanced filtering techniques.

In this mode, $BASELINE = 0$ and $ADCOUT = Raw\ ADC\ Value$ (in two's complement). Output reflects absolute sensor capacitance with electrical offset and mechanical preload included. AUTOCAL and INTRTHR thresholds become absolute thresholds (not baseline-referenced). To make preload adjustments manually, AUTOPRELDADJ should be disabled (set to 0) and then adjustments made through PRELDADJ register.

Register map

REGISTER	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0	ADDRESS	POR	R/W
CONFIGURATION & STATUS											
Main Configuration	TOEN	WAIT[2:0]			ADCRAW	TEMPWAIT[1:0]		EN	0x00	0XB5	R/W
Analog Configuration		INAGAIN[2:0]				PRECHARGE[2:0]			0x01	0x50	R/W
Status Register	BUSY				OVRINTRTH	OVRACALTH		INTR ¹	0x02	0x00	R/RCLR
ADC READING											
ADC Sensor High Byte	ADCOUT[15:8]								0x03	0x00	R
ADC Sensor Low Byte & Sample Counter-12bit	ADCOUT[7:4]				ADCOUT[3:0] / SCOUNT12[3:0]				0x04	0x00	R
Sample Counter-16bit	SCOUNT16[7:0]]								0x05	0x00	R
Misc. Control	AUTOPRELDADJ	ENNEGZONE	ENFILTER	HIGHORDER		FALLTHRSEL[2:0]			0x06	0x84	R/W
AUTOCAL DEFINITION											
Autocal Threshold High Byte	AUTOCAL[15:8]								0x07	0x02	R/W
Autocal Threshold Low Byte	AUTOCAL[7:0]								0x08	0x00	R/W
Autocal Config-1	ENCALMODE	CALRESET[2:0]			INTRPOL	CALPERIOD[2:0]			0x09	0XC2	R/W
Autocal Config-2		RISEBLWGT[2:0]				LIFTDELAY[2:0]			0x0A	0x20	R/W
Autocal Config-3	INVERT		TCO[1:0]			FALLBLWGT[2:0]			0x0B	0x02	R/W
INTERRUPT DEFINITION											
Interrupt Threshold High Byte	INTRTHR[15:8]								0x0C	0x00	R/W
Interrupt Threshold Low Byte	INTRTHR[7:0]								0x0D	0x00	R/W
Interrupt Configuration	INTREN	INTRMODE	INTRPERSIST	BTNMODE		INTRSAMPLES[2:0]			0x0E	0x03	R/W
MISCELLANEOUS											
Baseline Control								FORCEBL	0x0F	0x00	R/W
Neg Zone Config		NEGRESET[2:0]				RESETWGT[2:0]			0x10	0x50	R/W
Sensor Baseline High Byte	BASELINE[15:8]								0x12	0x00	R
Sensor Baseline Low Byte	BASELINE[7:0]								0x13	0x00	R
Neg Zone							NEGTHSEL[1:0]		0x14	0x00	R/W
Preload Adjust		PRELDADJ[6:0]							0x15	0x00	R/W
Falling Threshold High Byte	FALLTHR[15:8]								0x16	0x00	R/W
Falling Threshold Low Byte	FALLTHR[7:0]								0x17	0x00	R/W
Temp Sensor High Byte	TEMP[11:4]								0x18	0x00	R
Temp Sensor Low Byte	TEMP[3:0]				TCOUNT[3:0]				0x19	0x00	R
Temp Sensor FIFO High Byte	TEMPBUF[11:4]								0x1E	0x00	R
Temp Sensor FIFO Low Byte	TEMPBUF[3:0]				TBCOUNT[3:0]				0x1F	0x00	R
FIFO Sensor Data High Byte	ADCBUF[15:8]								0x2E	0x00	R
FIFO Sensor Data Low Byte	ADCBUF[7:4]				ADCBUF[3:0] / FCOUNT[3:0]				0x2F	0x00	R
Device and Revision ID	DEVID[4:0]					REV[2:0]			0x80	0x2A	R

¹The INTR bit is cleared immediately after I²C read of the STATUS register or register 0x01.

Register Definitions

REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x00	Reg00[7]	TOEN	1b1	0: I ² C time out disabled 1: I ² C time out enabled. Time out=51.2 ms typ, resets the bus if there is no activity.	R/W
0x00	Reg00[6:4]	WAIT[2:0]	3b011	Wait period between successive force sensor samples. 000: 0 ms 001: 1 ms 010: 4 ms 011: 8 ms 100: 16 ms 101: 32 ms 110: 64 ms 111: 256 ms	R/W
0x00	Reg00[3]	ADCRAW	1b0	Indicates baselined or raw sensor ADC conversion. 0: Baselined ADC sensor measurement. In this mode, the baseline is subtracted from the raw sensor measurement in 2's complement. 1: Raw ADC sensor measurement in 2's complement.	R/W
0x00	Reg00[2:1]	TEMPWAIT[1:0]	2b10	Number of sensor ADC samples between temperature conversion. When WAIT[2:0]=000, no temperature conversion can occur. 00: No temperature conversion 01: Convert every sample 10: Convert every 16 samples 11: Convert every 32 samples	R/W
0x00	Reg00[0]	EN	1b1	0: Disable sensor conversion, shutdown mode 1: Enable sensor conversion	R/W
0x01	Reg01[6:4]	INAGAIN[2:0]	3b101	Instrumentation amplifier gain selection. 000: x1 001: x8 010: x16 011: x31 100: x61 101: x115 110: x208 (set PRECHARGE ≥ 200 μs) 111: x416 (set PRECHARGE ≥ 400 μs)	R/W
0x01	Reg01[2:0]	PRECHARGE[2:0]	3b000	Sensor bridge settling time selection. The ADC will wait for this much time before conversion starts. 000: 50 μs 001: 100 μs 010: 200 μs 011: 400 μs 100: 800 μs 101: 1.6 ms 110: 3.2 ms 111: 6.4 ms	R/W
0x02	Reg02[7]	BUSY	1b0	Device boot in progress. ADC conversion is not accurate until boot is complete. 0: Boot done 1: Boot in progress	R
0x02	Reg02[3]	OVRINTRTH	1b0	The last ADC reading from sensor was above the interrupt threshold. 0: ADC reading under interrupt threshold 1: ADC reading over interrupt threshold	R

REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x02	Reg02[2]	OVACALTH	1b0	The last ADC reading from sensor was above the auto-calibration threshold. 0: ADC reading under auto-calibration threshold 1: ADC reading over auto-calibration threshold	R
0x02	Reg02[0]	INTR	1b0	Status bit to indicate new sensor interrupt. A read of the STATUS register clears this bit and the hardware interrupt (if BTNMODE=0 and INTRPERSIST=1). This bit is set and cleared even if the INTREN bit is low. 0: No interrupt 1: Interrupt generated	RCLR
0x03	Reg03[7:0]	ADCOUT[15:8]	8b00000000	Digitized 2's complement sensor ADC out.	R
0x04	Reg04[7:0]	ADCOUT[7:0]	8b00000000	ENFILTER = 0, use ADCOUT[15:4], -2048 to +2047 ENFILTER = 1, use ADCOUT[15:0], -32,768 to +32,767	
0x04	Reg04[3:0]	SCOUNT12[3:0]	4b0000	ADC sample counter for ULP mode. Wraparound sample counter increments by 1 after each sample. Used to assist data timestamping & algorithm post-processing. Associated with ADC data reading.	R
0x05	Reg05[7:0]	SCOUNT[7:0]	8b00000000	ADC sample counter for HP mode. Wraparound sample counter increments by 1 after each sample. Used to assist data timestamping & algorithm post-processing. Associated with ADC data reading.	R
0x06	Reg06[7]	AUTOPRELDADJ	1b1	Automatic offset adjustments. When enabled, the PRELDADJ[6:0] is automatically set to minimize offset. Whenever the baseline is above 512 ADC counts then it is reduced to below it and if the baseline is below -1024 ADC counts then it is increased to above it, using PRELDADJ. 0: Disabled 1: Enabled	R/W
0x06	Reg06[6]	ENNEGZONE	1b0	Allow negative baseline settings 0: Disabled 1: Enabled	R/W
0x06	Reg06[5]	ENFILTER	1b0	Enable 16-bit and averaging filter, applies to raw ADC data before baselining 0: Disabled 1: Enabled	R/W
0x06	Reg06[4]	HIGHORDER	1b0	Sets order of averaging filter when ENFILTER = 1 0: 4-point sum 1: 16-point sum	R/W
0x06	Reg06[2:0]	FALLTHRSEL[2:0]	3b100	Falling threshold when BTNMODE=1: Defined as fraction of falling threshold value (FALLTHR[15:0] register 0x16/0x17). 000: FALLTHR[15:0] X 1 Defined as fraction of interrupt threshold value (INTRTHR[15:0] register 0x0C/0x0D). 001: INTRTHR[15:0] X 0.25 010: INTRTHR[15:0] X 0.375 011: INTRTHR[15:0] X 0.5 100: INTRTHR[15:0] X 0.625 101: INTRTHR[15:0] X 0.75 110: INTRTHR[15:0] X 0.875 111: INTRTHR[15:0] X 1	R/W
0x07	Reg07[7:0]	AUTOCAL[15:8]	8b00000010	Auto-calibration threshold setting for sensor in 2's complement. Always use positive values (MSB first) and it shall be lower than Interrupt threshold value. In ULP mode, lower nibble is not used. When ENFILTER is set, all 16 bits are used for data.	R/W
0x08	Reg08[7:0]	AUTOCAL[7:0]	8b00000000		

REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x09	Reg09[7]	ENCALMODE	1b1	CALRESET trigger zone and weight selection. ENCALMODE = 0 is recommended for cases with very long press and ENCALMODE = 1 is recommended for button applications as it helps to recover a stuck condition. 0: CALRESET timeout triggers auto-calibration when sensor readings are between AUTOCAL threshold and INTRTHR threshold. Uses RESETWGT at intervals of CALRESET to recover. 1: CALRESET timeout triggers auto-calibration when sensor readings are above AUTOCAL threshold. Uses RESETWGT at intervals of CALRESET to recover.	R/W
0x09	Reg09[6:4]	CALRESET[2:0]	3b100	Defines timeout before triggering baseline recalibration when sensor measurement remain above auto-calibration threshold. Upon expiry of this timer, the new baseline value is calculated based on ENCALMODE and RESETWGT. 000: 0.5 s 001: 1 s 010: 2 s 011: 4 s 100: 8 s 101: 16 s 110: 32 s 111: Infinite (no reset, unless forced)	R/W
0x09	Reg09[3]	INTRPOL	1b0	Interrupt polarity. 0: INTB pin active low (recommended) 1: INTB pin active high	R/W
0x09	Reg09[2:0]	CALPERIOD[2:0]	3b010	Auto-calibration timeout. Defines the time interval for auto-calibration operation when the sensor measurement is below auto-calibration threshold. 000: 25 ms 001: 50 ms 010: 100 ms 011: 200 ms 100: 400 ms 101: 800 ms 110: 1600 ms 111: Infinite (no recalibration, unless forced)	R/W
0x0A	Reg0A[6:4]	RISEBLWGT[2:0]	3b010	Defines the weight of the current baseline to compute the new baseline on CALPERIOD expiry, when the sensor measurement is higher than current baseline. Higher values make the baseline change slower. $((\text{current baseline})(\text{RISEBLWGT}) + \text{current ADC reading})/(\text{RISEBLWGT} + 1)$ 000: 0 001: 1 010: 3 011: 7 100: 15 101: 31 110: 63 111: 127	R/W

REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x0A	Reg0A[2:0]	LIFTDELAY[2:0]	3b000	Defines the wait period before resuming baselining after the sensor value falls below the auto-calibration threshold following an interrupt threshold exceedance. 000: 0 ms 001: 20 ms 010: 40 ms 011: 80 ms 100: 160 ms 101: 320 ms 110: 640 ms 111: 460 ms	R/W
0x0B	Reg0B[7]	INVERT	1b0	Digital signal inversion 0: Not inverted 1: Inverted	R/W
0x0B	Reg0B[5:4]	TCO[1:0]	2b00	Sensor temperature coefficient of offset adjustment 00: 0 01: -20uV/V/°C 10: -40uV/V/°C 11: -60uV/V/°C	R/W
0x0B	Reg0B[2:0]	FALLBLWGT[2:0]	3b010	Defines the weight of the current baseline to compute the new baseline on CALPERIOD expiry, when the sensor measurement is lower than current baseline. Higher values make the baseline change slower. $((\text{current baseline})(\text{FALLBLWGT}) + \text{current ADC reading}) / (\text{FALLBLWGT} + 1)$ 000: 0 001: 1 010: 3 011: 7 100: 15 101: 31 110: 63 111: 127	R/W
0x0C	Reg0C[7:0]	INTRTHR[15:8]	8b00000000	Interrupt threshold setting for sensor in 2's complement when signal is rising. Default value corresponds to the resistance on the THR_ADR pin. Always use positive values (MSB first). In ULP mode, lower nibble is not used. When ENFILTER is set, all 16 bits are used for data. Do not set INTRTHR lower than AUTOCAL.	R/W
0x0D	Reg0D[7:0]	INTRTHR[7:0]	8b00000000		
0x0E	Reg0E [7]	INTREN	1b0	Determines if the INTB pin is driven with the sensor measurements. Does not mask the INTR status bit. Clear this bit (set to 0) before exiting shutdown (EN set to 1). If needed, re-enable no earlier than $\Delta t = 3 * (\text{WAIT} + \text{PRECHARGE} + 1)$ after setting EN to 1. 0: INTB pin not driven upon interrupt, unless BTNMODE = 1 1: INTB pin driven upon interrupt	R/W
0x0E	Reg0E[6]	INTRMODE	1b0	Interrupt mode. Controls if interrupt is generated by each sensor ADC conversion or after meeting the INTRSAMPLES requirement. 0: Uses INTRSAMPLES to qualify interrupt 1: Interrupt generated after each sensor ADC conversion if the sample value is above interrupt threshold	R/W
0x0E	Reg0E[5]	INTRPERSIST	1b0	Sets how long interrupt is pulled low. 0: 100 μ s interrupt pulse 1: Infinite, until the Status register is read	R/W

REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x0E	Reg0E[4]	BTNMODE	1b0	Enable button mode. Clear this bit (set to 0) before exiting shutdown (EN set to 1). If needed, re-enable no earlier than $\Delta t = 3 * (\text{WAIT} + \text{PRECHARGE} + 1)$ after setting EN to 1. 0: Disable button mode, see "Interrupt Modes" section for detail. 1: Enable button mode, falling threshold to de-assert interrupt.	R/W
0x0E	Reg0E[2:0]	INTRSAMPLES[2:0]	3b011	Number of samples above interrupt threshold after which an interrupt is asserted (and de-asserted when BTNMODE=1). 000: 1 001: 1 010: 2 011: 3 100: 4 101: 5 110: 6 111: 7	R/W
0x0F	Reg0F[0]	FORCEBL	1b0	Force baseline (ADC reading transferred to Baseline). 0: Auto-Baseline enabled 1: Forces baseline to the next sensor ADC reading. This bit is automatically reset after the baseline is updated.	R/W
0x10	Reg10[6:4]	NEGRESET[2:0]	3b101	Define timeout before triggering baseline recalibration. Baseline recalibration occurs when sensor measurements remain below the negative auto-calibration threshold for the timeout period. The new baseline is weighted based on RESETWGT[2:0] setting. 000: 0.5 s 001: 1 s 010: 2 s 011: 4 s 100: 8 s 101: 16 s 110: 32 s 111: No reset	R/W
0x10	Reg10[2:0]	RESETWGT[2:0]	3b000	Define the weight of the current baseline to compute the new baseline when baseline update is a result of a CALRESET or NEGRESET timeout. $((\text{current baseline})(\text{RESETWGT}) + \text{current ADC reading})/(\text{RESETWGT} + 1)$ 000: 0 001: 1 010: 3 011: 7 100: 15 101: 31 110: 63 111: 127	R/W
0x12	Reg12[7:0]	BASELINE[15:8]	8b00000000	Digitized 2's complement sensor baseline data, ENFILTER = 0, use BASELINE [15:4], -2048 to +2047	R
0x13	Reg13[7:0]	BASELINE[7:0]	8b00000000	ENFILTER = 1, use BASELINE [15:0], -32,768 to +32,767	
0x14	Reg14[1:0]	NEGTHSEL[1:0]	2b00	Sets negative autocalibration threshold level as a fraction of AUTOCAL[15:0] in conjunction with ENNEGZONE. 00: X(-1) 01: X(-0.5) 10: X(-0.25) 11: X(-0.125)	R/W

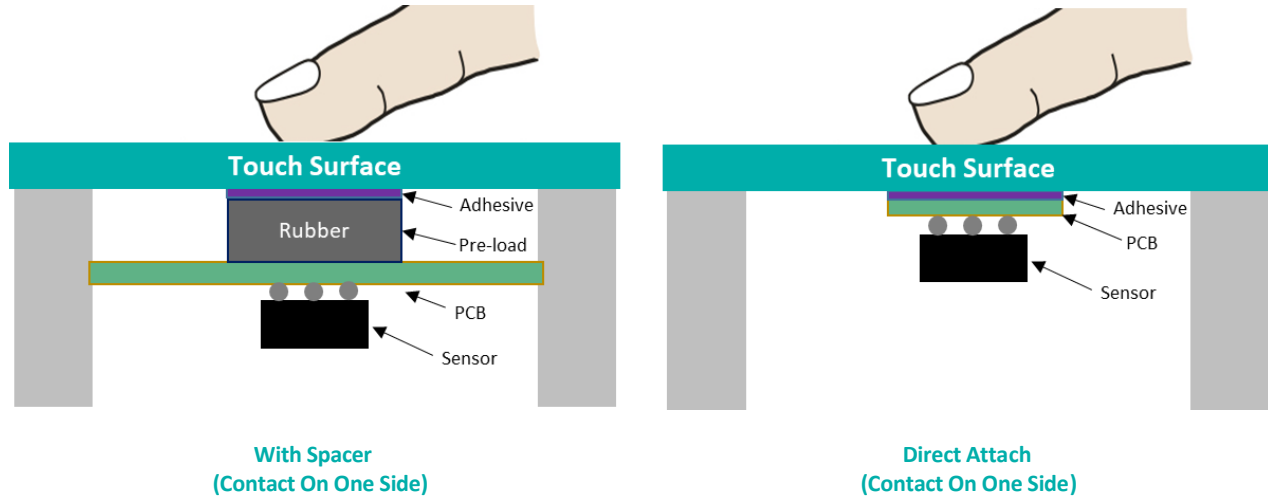
REGISTER ADDRESS	DATA BITS	BIT FIELD NAME	DEFAULT	DESCRIPTION	READ/ WRITE
0x15	Reg15[6:0]	PRELDADJ[6:0]	7b0000000	7-bit sensor offset adjustment; MSB is sign bit. It is read-only when AUTOPRELDADJ = 1. 3.5 Counts * GAIN / LSB	R/W
0x16	Reg16[7:0]	FALLTHR[15:8]	8b00000000	Interrupt threshold setting to de-assert INTB for sensor in 2's complement when signal is falling, BTNMODE=1, and FALLTHRSEL[2:0]=000. In ULP mode, lower nibble is not used. When ENFILTER is set, all 16 bits are used for data.	R/W
0x17	Reg17[7:0]	FALLTHR[7:0]	8b00000000	Only writable when FALLTHRSEL[2:0]=000, read-only otherwise.	
0x18	Reg18[7:0]	TEMP[11:4]	8b00000000	Digitized 2's complement temperature sensor data. Divide the 12-bit number by 16 to get the temperature in °C.	R
0x19	Reg19[7:4]	TEMP[3:0]	4b0000		
0x19	Reg19[3:0]	TCOUNT[3:0]	4b0000	Temperature sample counter. Wraparound sample counter increments by 1 after each sample. Used to assist data timestamping & algorithm post-processing. Associated with temperature data reading.	R
0x1E	Reg1E[7:0]	TEMPBUF[11:4]	8b00000000	Digitized 2's complement temperature sensor buffered data (16 sample deep).	R
0x1F	Reg1F[7:4]	TEMPBUF[3:0]	4b0000	Burst read the entire buffer before requesting another register read command. When the I2C read pointer reaches this register addresses, it will just loop back onto itself.	
0x1F	Reg1F[3:0]	TBCOUNT[3:0]	4b0000	Buffered temperature sample counter.	R
0x2E	Reg2E[7:0]	ADCBUF[15:8]	8b00000000	Digitized 2's complement sensor buffered ADC out. Reading the Buffer more frequently than 300µs apart could result in reading the same data, since the internal clock is 10kHz which is used to manage the read/write logic. The Buffer data and FCOUNT read back 0x0000 when all the data stored has been read back.	R
0x2F	Reg2F[7:0]	ADCBUF[7:0]	8b00000000	Burst read the entire buffer before requesting another register read command. When the I2C read pointer reaches this register addresses, it will just loop back onto itself.	
0x2F	Reg2F[3:0]	FCOUNT[3:0]	4b0000	ADC buffered 4-LSB sample counter. The Buffer data and FCOUNT read back 0x0000 when all the data stored has been read back.	R
0x80	Reg80[7:3]	DEVID[4:0]	5b00101	Identifier for sensor	R
0x80	Reg80[2:0]	REV[2:0]	3b010	Revision identifier	R

Sample Code

I ² C WRITE	I ² C READ
<pre> u8 ni_i2c_write(struct ni_sensor_t s, u8 reg, u8 length, u8 *data) { u8 error = NI_OK; //ATTENTION: FOLLOWING CODE SNIPPET IS AN EXAMPLE //FOR ATMEL STUDIO, USER SHOULD CHANGE IT TO //WORK ON THEIR PLATFORM. u8 i; u8 status; u8 buffer[length+2]; buffer[0] = s.addr << 1; buffer[1] = reg; for(i=0; i<length; i++) { buffer[i+2] = data[i]; } status = USI_TWI_Start_Transceiver_With_Data(buffer, length+2); USI_TWI_Master_Stop(); if (status == 0) { error = NI_ERROR_COM; } //END CODE SNIPPET return error; } </pre>	<pre> u8 ni_i2c_read(struct ni_sensor_t s, u8 reg, u8 length, u8 *data) { u8 error = NI_OK; //ATTENTION: FOLLOWING CODE SNIPPET IS AN //EXAMPLE FOR ATMEL STUDIO, USER SHOULD //CHANGE IT TO WORK ON THEIR PLATFORM. u8 i; u8 status; u8 buffer[length+1]; buffer[0] = s.addr << 1; buffer[1] = reg; //USI_TWI_Start_Transceiver returns 0 if unsuccessful status = USI_TWI_Start_Transceiver_With_Data(buffer, 2); if (status == 0) { error = NI_ERROR_COM; return error; } //Read from registers buffer[0] = (s.addr << 1) 1; status = USI_TWI_Start_Transceiver_With_Data(buffer, length+1); USI_TWI_Master_Stop(); if (status == 0) { error = NI_ERROR_COM; return error; } //Copy read values from buffer into data for(i=0; i<length; i++) { data[i] = buffer[i+1]; } //END CODE SNIPPET return error; } </pre>

Solution Stack-up

The sensor is soldered onto a PCB or FPCB, which is then bonded to the back of the surface where the force is to be detected. Below figure shows the stack-up for a single button application utilizing one sensor mounted to a PCB. For substrate options and recommended adhesives, please contact Semtech.

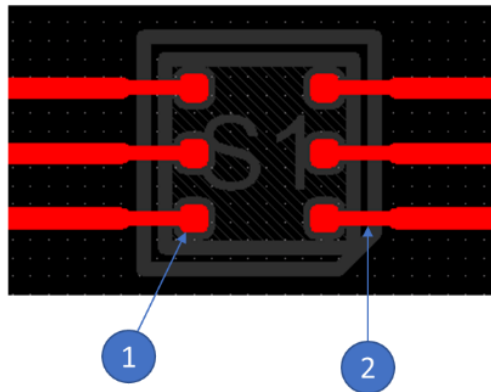


Electrical Design, Layout, Placement, Handling and Package Information

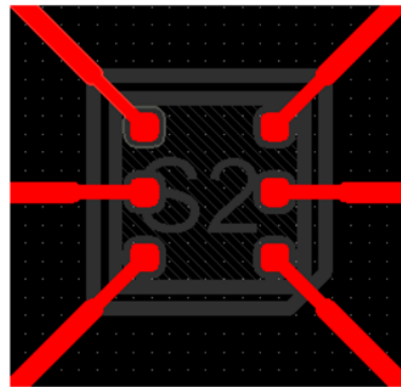
Electrical Design and Board Layout Considerations

- Two bypass capacitors are required, 2.2 μ F & 1 nF, to reduce noise on the power supply. Use higher value bypass capacitors for noisy power supplies. Ensure that both capacitors are placed close to the sensor – **REQUIRED**.
- Symmetrical routing improves sensor self-alignment during assembly – **REQUIRED**.
- Place the nearest components at least 0.3 mm (RECOMMENDED) away from the sensor.
- Do not use vias directly under the sensor solder balls.
- Keep signal lines short (**REQUIRED**) and free of 90° turns (steep angles/corners can cause undesired acid traps during the manufacturing process). Use 45° turns or rounded-edge-turns for all signal/power lines.
- Do not use glue/epoxy between sensor and board or around the sensor – **REQUIRED**.
- The sensor edge should be at least 0.25 mm from all edges of the PCB – **RECOMMENDED**.
- If using a flexible PCB, a top hatched ground plane is recommended.
- Underfill is not recommended. For harsh environments like salt spray please contact Semtech for recommendations for materials for under-fill and glob top.
- For additional stability consider implementing “teardrop” routing into vias, pin pads and T-junctions; this increases the copper area and avoids steep angles in areas where multiple traces meet in a common point.
- Power the AFE from a dedicated on-board low-noise LDO regulator to minimize supply-induced noise coupling.
- Make the traces to all the pins the same width. Semtech recommends greater than (or equal) 6 mils traces for all signals except the traces directly connected to bumps of the sensor. These should be 4 mil traces for better reflow results.

Side Entry

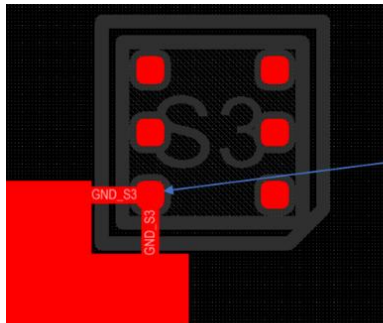


Corner Entry

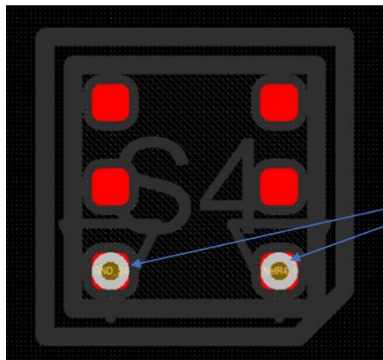


- 1 → Pad Size: 0.2mm square pad with rounder corner.
Solder Mask: 0.05mm wider on all four sides of pad.
Copper defined, not solder mask defined pad.
- 2 → Use 0.1mm trace-width (1/2 of pad size) at pad entry for minimum of 0.25mm length.
After 0.25mm: can increase trace width to 0.15mm.

Recommended Design/Layout Practice



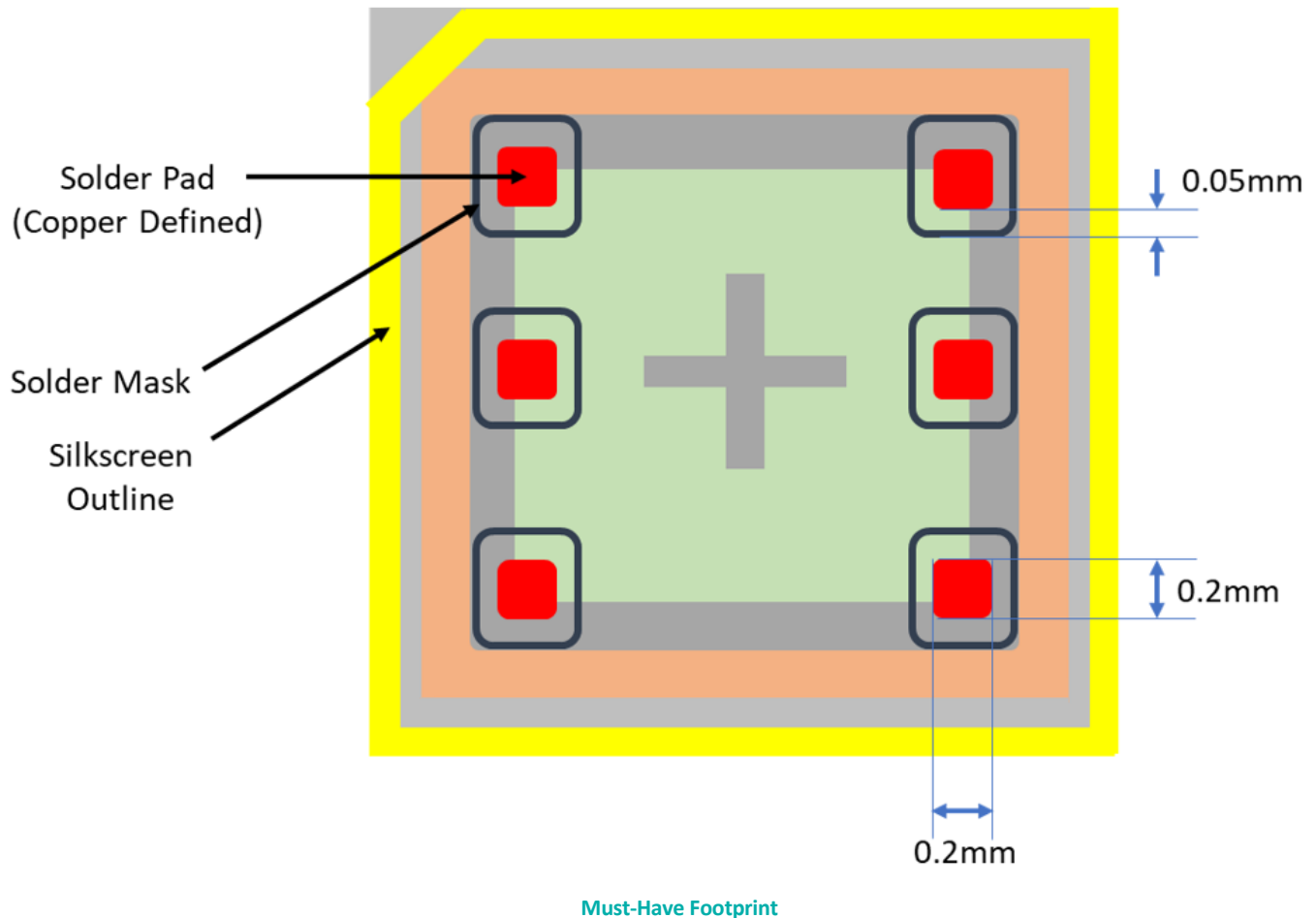
Do not use multiple traces on same pads.
Do not use copper pouring or plane direct on device pad.
This will make routing non-symmetrical with other pads.



Do not use Via in pad or micro via (Blind via) direct on land area.

NOT Recommended Design/Layout Practice

Footprint and SMT Details



Must-Have Footprint

Figure above depicts the must-have footprint drawing for the sensor and the details below must be followed for the SMT of the sensor onto a PCB/FPCB

- Pad dimensions and type
 - Rounded rectangle with dimensions: **7.874 mil x 7.874 mil (0.2 mm x 0.2 mm)**
 - Radius of the rounded square pad: **corner radius of 0.05 mm (not a circular pad)**
 - Copper defined, **not solder-mask defined**
- Solder mask: **2 mil (50 µm) wider on all sides of the pad.**
- Solder paste stencil thickness: **3 mil (76 µm)**
- Solder paste stencil aperture: **1:1 with the copper pad (0.2 mm x 0.2 mm)**
- Solder Paste: **Alpha OM 550 HRL1 (to reduce solution offset and TCO), SAC305 or SAC405**

Device Handling – Best Practices Design

- Do not pick up the device with metal tweezers. Use a vacuum pickup tool.
- Do not “snap” panelized, assembled PCBs
- Follow ESD-safe handling recommendations
 - Store sensors in ESD containers (e.g. T&R, moisture-sealed)
 - Handle devices only in ESD-safe work areas
 - People /machines handling the sensor must be grounded to avoid potential ESD damage

SMT Pick and Place Guidelines

Nozzle Considerations

SMT assembly machines typically employ a vacuum nozzle for Pick & Place (P&P) operations. Not all equipment is the same, but there are some general guidelines to follow, for optimal performance/yield.

- Select a vacuum nozzle only.
 - DO NOT use mechanical grippers or collets that contact the edges/sides of a WLCSP.
 - DO NOT handle WLCSP parts with metal tweezers.
- Select softer nozzle materials where possible.
 - Nozzle materials can vary (metal, ceramic, plastic, rubber), and certain equipment requires certain nozzle materials. Select softer material to avoid potential damage from mechanical shocks.
- Select an appropriate nozzle tip shape.
 - Nozzle tip shape can affect the amount of pressure applied to a single point on a WLCSP part.
 - Nozzle tips should be circular, square/rectangular, and preferably with a vertical separator.
 - The nozzle tip surface should be a single plane; do not use a nozzle with a protruded edge.



Acceptable Vs NOT acceptable Tip Shapes

- Select an appropriate nozzle tip size, for the part being placed.
 - The largest tip dimension should be smaller than the size of the WLCSP part.
 - The tip should account for the tolerance of the pick accuracy, so that it's not possible for the nozzle tip to contact the edge or the corner of a part.
 - The tip size should be as large as possible, while satisfying the above 2 points.

Pick and Place Operation

The P&P operation consists of two key steps: picking a part out of the carrier tape pocket, and placement of that part onto a circuit board. Both operations are performed with some alignment tolerance: nozzle-to-part tolerance during the pick, and part-to-board alignment during the placement. Additionally, the placement should occur with as little force as possible, to press the die into place on the board.

WLCSP parts are shipped to customers in standard carrier tape & reel packaging. Parts sit in a pocket that is slightly larger than the size of the part. A part can move laterally inside the pocket, typically by less than 0.1 mm, but this varies by product.

- If available, the P&P equipment should use a vision detection system to align the P&P nozzle to the center of the die, for every part picked.
- If a vision alignment system isn't available, the nozzle size selection should account for the positional tolerance of the WLCSP part in the carrier tape pocket.

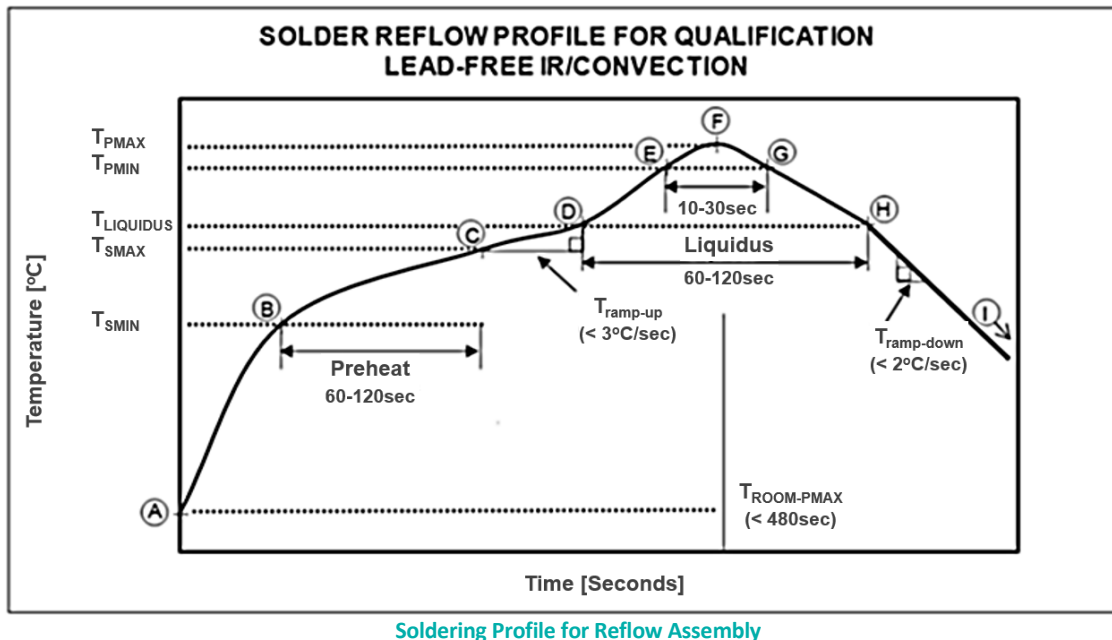
The die placement operation should be performed with a placement accuracy of ≤ 0.1 mm; placement accuracy of ≤ 0.05 mm is typical with modern P&P equipment. The placement force shall be the minimum required to bring the WLCSP solder bumps into contact with the solder paste on the board.

A contactless vacuum pickup is strongly preferred; the vacuum force alone should lift the part with only incidental surface contact to the die top.

- If air placement isn't available, use a contact mode for the part placement. Ensure that both the pick and placement force is not larger than 2 N (~200 gf). Placement forces should be verified periodically using a calibrated load cell, to ensure parts aren't placed with a force over this specification.

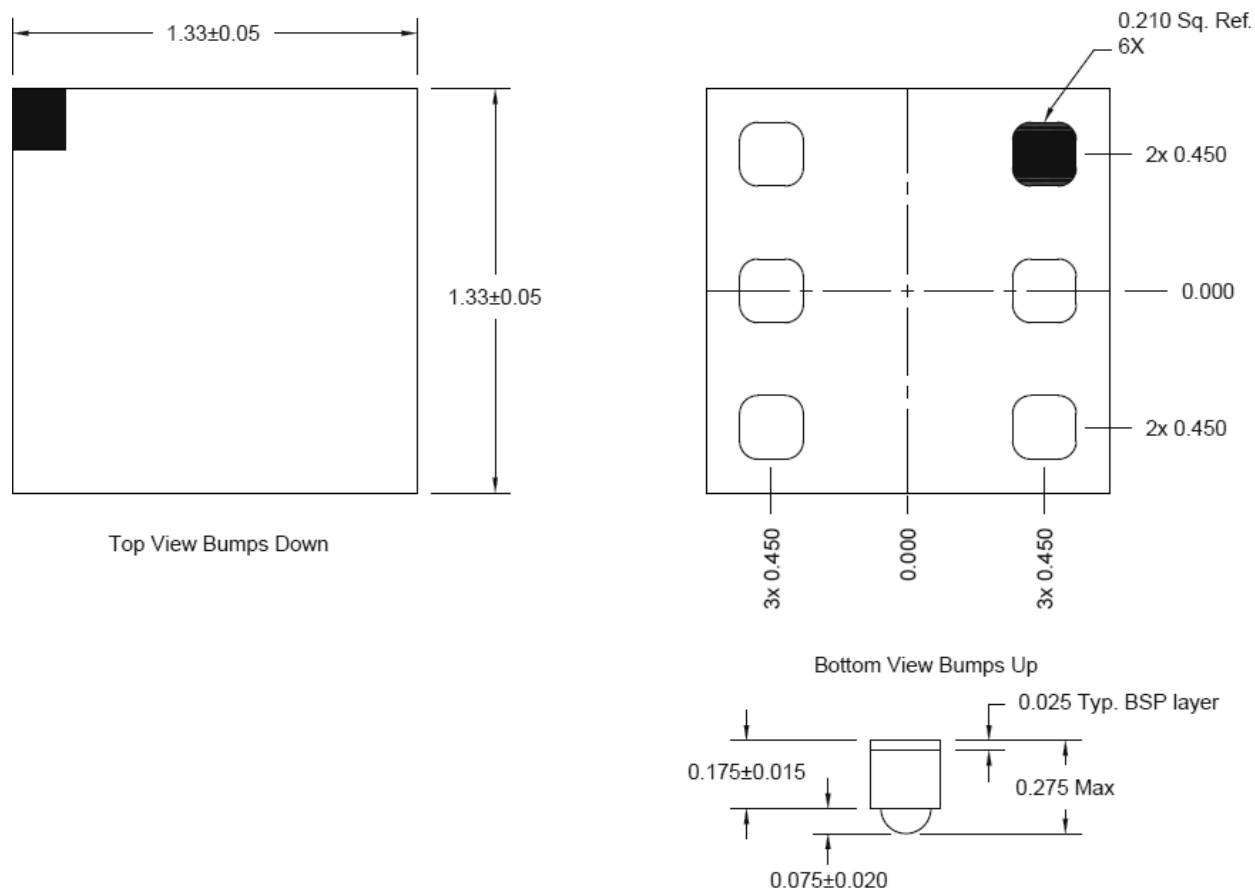
Assembly Instructions

The chip is designed to be reflow-soldered using direct-chip attachment (DCA) techniques to the circuit substrate (e.g. FR-4 or FPCB) and reflow profile that is designed for RoHS and Pb-free compliance (see the profile below). Assembly houses should follow this profile closely but can choose more conservative ramp-up/down rates. To avoid damage to the chip do not exceed the maximum ratings of the qualification profile (e.g. T_{PMAX} = 260 °C, measured at the top side of the WLCSP). Customers must verify with their assembly house for the appropriate reflow profile.



STEP	PARAMETER	TEMPERATURE [°C]	TIME [S]	MAXIMUM RATE [°C/S]
A	T _{ROOM}	25		
B	T _{SMIN}	150		
C	T _{SMAX}	200	60 < t _{BC} < 120	
D	T _{LIQUIDUS}	217		r (T _{PMAX} - T _{LIQUIDUS}) < 3
E	T _{PMIN} [255 °C, 260 °C]	255		r (T _{PMAX} - T _{LIQUIDUS}) < 3
F	T _{PMAX} [260 °C, 265 °C]	260	t _{AF} < 480	r (T _{PMAX} - T _{LIQUIDUS}) < 3
G	T _{PMIN} [255 °C, 260 °C]	255	10 < t _{EG} < 30	r (T _{PMAX} - T _{LIQUIDUS}) < 2
H	T _{LIQUIDUS}	217	60 < t _{DH} < 120	
I	T _{ROOM}	25		

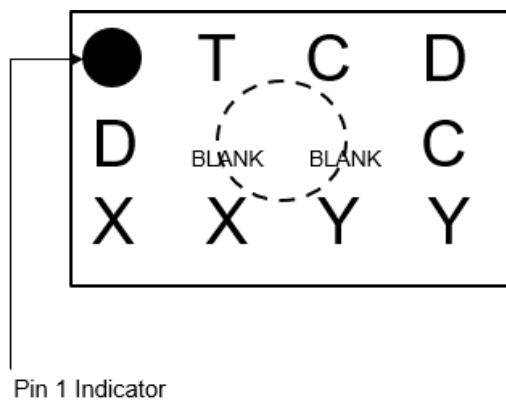
Mechanical Information



All dimensions are in mm.

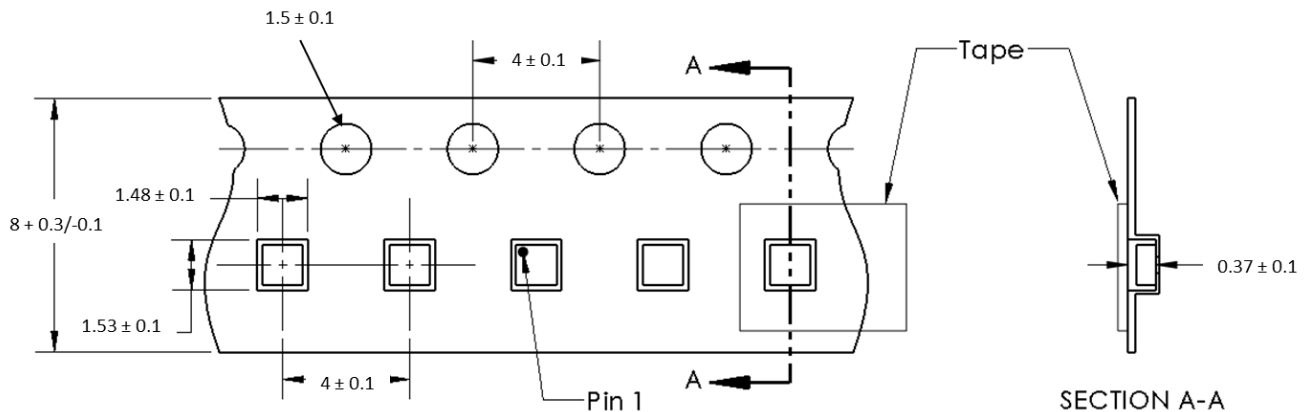
Product Marking

The product marking consists of 3 rows of letters and numbers printed on the top of WLCSP package.



Product Marking - Line1: Trace Code, Line2: Product ID, Line3: Die Traceability

Tape and Reel Information



All dimensions and tolerances in this tape and reel diagram are in mm.

The image below depicts a tape reel sealed in an ESD-protective bag. The diameter of the sensor reel measures 7 inches (Radius = 3.5 inches) with reel thickness of 0.25 inches to comfortably host the tape. Each reel is stamped with the company logo, part number, lot number, and date code.



Reliability and Environmental Information

The Semtech sensor meets Level 1 (unlimited) Moisture Sensitivity Level (MSL) specifications.

Reliability and Environmental reports are furnished upon request.

GLOSSARY (Alphabetical Order)

CDM (Charge Device Model) – CDM is a model for characterizing the susceptibility of an electronic device to damage from electrostatic discharge (ESD). CDM is commonly used for semiconductor devices and has been standardized in JEDEC with JEDEC JS-002.

HBM (Human Body Model) – HBM is the most used model for characterizing the susceptibility of an electronic device to damage from electrostatic discharge (ESD). The model simulates an ESD discharge that may occur when a human contacts an electronic device.

I²C Timeout – This is the maximum time period during which the I²C bus can remain idle before it automatically resets.

Revision History

REVISION NUMBER	REVISION DATE	DESCRIPTION/CHANGES	PAGES CHANGED
1	04/17/2026	New Release	NA



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